

☆☆☆☆☆ IRONHACK AI ENGINEERING · PROJECT 4

What 34,660 reviews really say

NLP Automated Customer Reviews: sentiment, product categories and
AI-written buyer's guides

Sentiment

Clustering

Generative AI

Deployment

Hamza Shabbir & Yasaman Najafi Jozani

One dataset, four tasks

OUR HYPOTHESIS

Review text alone is enough to detect sentiment, group products and write trustworthy buyer's guides.

Kaggle "Consumer Reviews of Amazon Products" ·
34,660 reviews · Kindle, Fire and Echo devices



Sentiment

negative, neutral or positive



Clustering

40 products, 5 groups



Summaries

a buyer's guide per group



Deployment

a web app anyone can try

Real-world data, real-world mess

34,660 → **27,863** clean reviews kept (80%)

11,299

glued names

"Echo,,,Charger": keep the first. 48 → 40 products

6,760

no product name

20% of rows, dropped

1 id

many products

clean name becomes the product key

1 file

shared cleaning

same text cleaning in training and in the app

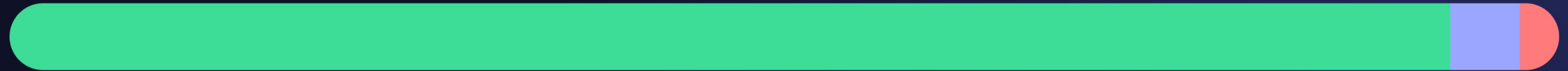
No APIs or scraping: the Kaggle file was enough. We enriched it with pretrained Hugging Face models.

03 · EXPLORATORY DATA ANALYSIS

93%

of reviews are positive

Every product averages 4.4–4.9 stars. Only 2% of reviews are negative, so accuracy alone would fool us.



Positive 93%

Neutral 4.5%

Negative 2.5%



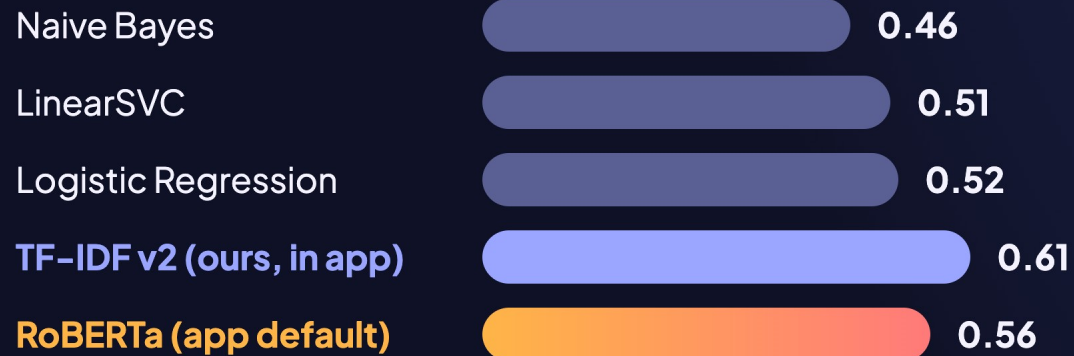
3 products hold 63% of all reviews



Top complaint words: "slow", "returned", "work"

Five models, one fair test

Macro-F1 on 8,359 test reviews (higher is better)



WHY ROBERTA IS DEFAULT

9 / 9

hand-written reviews right (TF-IDF: 5 / 9)

Finds 80% of negative reviews (TF-IDF: 51%)

Both models are in the app. Weak spot: mixed 3-star reviews. Macro-F1 because "always positive" already scores 93% accuracy.

05 · TASK 4 · THE APP · LIVE DEMO

Paste a review, pick a model

</> Streamlit app: emoji label + 3 confidence bars

⚙️ Model picker: RoBERTa or our TF-IDF v2

🗄️ Both models live on Hugging Face, not in Git

☁️ GitHub → Streamlit Community Cloud

cardiffnlp/twitter-roberta-base-sentiment-latest

hamza-shabbir94/amazon-review-sentiment-analysis

★ Amazon Review Sentiment

Paste a product review and get negative / neutral / positive. Both models are hosted on Hugging Face.

Model

☒ RoBERTa (pretrained) ☐ TF-IDF v2 + LogReg (our model)

Hugging Face repo: [cardiffnlp/twitter-roberta-base-sentiment-latest](#)

Try an example (optional)

Great tablet for the price. My kids love it and the battery lasts all day.

Customer review

Great tablet for the price. My kids love it and the battery lasts all day.

Analyse sentiment

😊 **Positive**

confidence 99% · RoBERTa (pretrained)

😊 Positive: 99%

😊 Neutral: 1%

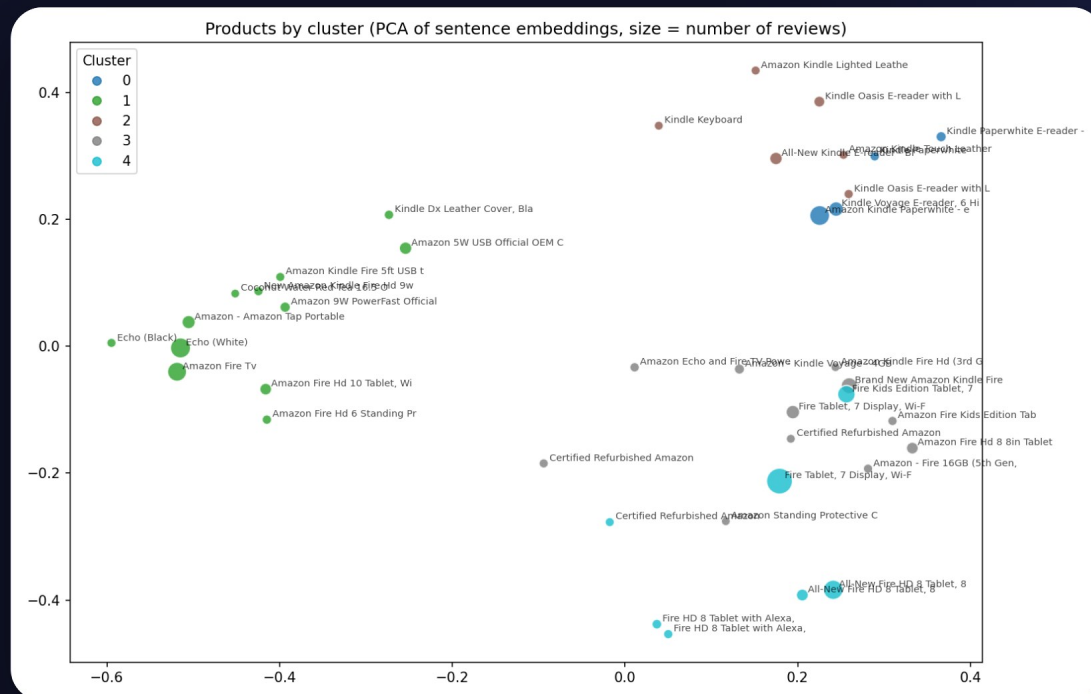
😊 Negative: 0%

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The live app: RoBERTa → 😊 Positive, 99%

06 · TASK 2 · PRODUCT CLUSTERING

40 products, 5 clusters



Name + category tags → MiniLM embeddings → KMeans, k = 5

0	Kindle e-readers	4 · 3,795
1	Echo, smart home, chargers	12 · 6,555
2	Kindle Oasis & covers	6 · 293
3	Fire 7" & older tablets	11 · 1,590
4	Fire HD 8 & Kids tablets	7 · 15,630

products · reviews

3 big families (e-readers, tablets, Echo). Silhouette ≈ 0.2 , so we checked each cluster by hand.

One buyer's guide per cluster



Two branches, one pipeline



Hamza

project structure, sentiment model, deployment



Yasaman

preprocessing, data cleaning, first summaries



How we worked

A GitHub branch each, merged by pull request. One notebook per task.



Plan vs reality

Kept the 4 tasks. Added a model comparison and redid clustering.



Risk management

Pinned versions, saved every step, two models to pick in the app.

Next time: one shared Python environment from day one (we lost time to 3.9 vs 3.13).

Our model passed the test and failed real users



"Worst customer service I have ever experienced" → **positive?!**

Why

Only ~570 negative training reviews, and cleaning deleted "not", "no", "too".

Lesson

Test with your own sentences, not only metrics. Check what cleaning removes.

In hindsight

Build the demo on day one and try a pretrained model early.

Hypothesis: mostly supported

Supported

Detect sentiment

Clear opinions right 9 of 9. Mixed 3-star reviews are still hard.

Partly

Find categories

3 clear groups, but weak separation (silhouette ≈ 0.2) and names set by hand.

Partly

Write guides

Real numbers via a facts sheet, but too long and some quotes misplaced.

Text works best when the question is clear and the **facts come before the AI writes.**

What surprised us, what's still open



Surprise

A pretrained model with zero training beat our own models on negative reviews.



Use case

With 93% positive reviews, a shop can auto-flag the few complaints that matter.



Still open

- Would fine-tuning close the gap?
- Beyond Amazon devices?



Thank you

Questions? Leave us a review.

Hamza Shabbir & Yasaman Najafi Jozani

[GitHub repo link]

Try the app: project-brief-nlp-automated-customers-reviews-sbmev8fypayseged.streamlit.app

Model: huggingface.co/hamza-shabbir94/amazon-review-sentiment-analysis